Naga Venkata Sai Nadh Vaka

M.Eng. Student | Embedded Systems • Electronics • Semiconductors

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SUMMARY

Master's student in Electrical & Microsystems Engineering at OTH Regensburg with a background in Electronics & Communication Engineering. Experienced in Embedded Systems, Electronics, Testing & Validation, and technical problem solving through research, engineering projects, and an Embedded Systems internship. Skilled in Python, C, MATLAB, microcontrollers, and laboratory measurements. Currently seeking Working Student and Internship opportunities in Embedded Systems, Semiconductor Technologies, Electronics Development, Testing & Validation, and related engineering fields.

EDUCATION

03/2026 – Present
Regensburg, Germany

Master of Engineering in Electrical and Microsystems
Ostbayerische Technische Hochschule (OTH) Regensburg
- Relevant Coursework: Advanced Semiconductor Technology, Electronic Product Engineering, Electronic Circuits & Systems, Applied Optics, Advanced Optoelectronics.

12/2021 – 04/2025
Vijayawada, India

Bachelor of Technology in Electronics and Communication Engineering
Usha Rama College of Engineering and Technology
- Graduated with Distinction (CGPA: 8.85/10; German Grade Equivalent: 1.6).
- President, ECRATZ ECE Student Association and Class Representative for 3 years.
- Qualified GATE 2024 (ECE).

EXPERIENCE

11/2024 – 04/2025
Vijayawada, India

Embedded Systems with IoT Applications Intern
Protective Technologies Pvt. Ltd.
- Supported development and testing activities for embedded systems using microcontrollers and sensor interfaces.
- Assisted in system validation, troubleshooting, and integration activities for embedded and IoT-based applications.
- Contributed to technical documentation, engineering workflows, and collaborative project support.

PROJECTS

Advanced Image Encryption Algorithm Integrating Chaotic Image Encryption and CNN
Final Year Research Project
- Implemented a hybrid image encryption framework using MATLAB and CNN-inspired feature extraction techniques.
- Analyzed system performance using PSNR, SSIM, NPCR, and histogram-based evaluation metrics.
- Co-authored and published research findings in a peer-reviewed engineering journal and communicated technical results through formal documentation.

Transformerless AC-to-DC Power Supply (5V/9V/12V/15V)
- Designed and tested a transformerless power supply prototype.
- Performed troubleshooting, validation, and performance evaluation using laboratory equipment.

SKILLS

- **Programming:** Python, C, MATLAB
- **Embedded & Electronics:** Embedded Systems, Electronics, Microcontrollers, IoT
- **Testing & Validation:** System Testing, Validation, Troubleshooting, Technical Documentation
- **Laboratory & Testing:** Oscilloscope, Digital Multimeter, DC Power Supply, Electronic Measurements
- **Software & Development Tools:** Git, Keil, Xilinx Vivado, Multisim, Proteus, Arduino IDE, MS Office
- **Languages: German** (A2 – actively improving), **English** (C1 – Fluent)
- **Professional:** Analytical Thinking, Communication, Teamwork, Problem Solving